

# WIDS - Accio Alpha

Santhosh BV 24B1290

---

## 1 Probability Theory

Probability is the measure of possibility of occurrence of an Event which belongs to a Sample Space. Here, Sample Space ( $\Omega$ ) is the set of all possible outcomes of an experiment and an Event is a subset of the sample space. This measure has certain properties as listed below

- **Non-Negative** The probability measure is non negative for all the events
- **Normalisation**  $P(\Omega) = 1$
- For mutually exclusive events  $A_1, A_2, \dots$   
 $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

### Some Properties

- **Complement Rule:**  $P(A^c) = 1 - P(A)$
- **Addition Rule:**  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Empty Set:**  $P(\emptyset) = 0$

**Conditional Probability** This is also a valid probability measure which conditions the occurrence of events on another event. It gives as the possibility of occurrence of an event given another event has occurred. This measure is widely used and plays a pivotal role in estimators. If  $P(B) > 0$ , the conditional probability of  $A$  given  $B$  is defined as

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

**Total Probability Principle** Let  $B_1, B_2, \dots, B_n$  be disjoint events whose union is the sample space. Then,

$$P(A) = \sum_{i=1}^n P(A | B_i) P(B_i)$$

**Bayes' Rule** For events  $A$  and  $B$  with  $P(B) > 0$ ,

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

**Independence of Events** Two events  $A$  and  $B$  are said to be independent if occurrence of one event does not depend on the occurrence of the other. This can be exhibited mathematically using the definitions of conditional probability as

$$P(A \cap B) = P(A) P(B)$$

Equivalently,

$$P(A | B) = P(A) \quad \text{and} \quad P(B | A) = P(B)$$

### Random Variables

Random Variables are functions which maps events from the sample space to real numbers. A random variable  $X$  is defined as  $X : \Omega \rightarrow R$ . Based on the range of values taken by the random variable it is divided into Discrete(Countable range) and Continuous random Variable(uncountable range).

Now in order to assign the probability measure to the random variables, the probability distribution of a borel set in the range of the random variable is defined as the probability of the set of events which results in the value of random variables in that set. This definition is used to determine the Probability Mass function for discrete Random variables and the Cumulative distribution function for continuous random variables.

### Probability Mass Function (PMF)

For a discrete random variable  $X$ , the probability mass function is defined as

$$p_X(x) = P(X = x)$$

#### Properties:

- $p_X(x) \geq 0$  for all  $x$
- $\sum_x p_X(x) = 1$  (sum over all possible values of  $X$ )

### Probability Density Function (PDF)

For a continuous random variable  $X$ , probabilities are described using a density function  $f_X(x)$  such that

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

#### Properties:

- $f_X(x) \geq 0$  for all  $x$
- $\int_{-\infty}^{\infty} f_X(x) dx = 1$

### Cumulative Distribution Function (CDF)

The cumulative distribution function of a random variable  $X$  is defined as

$$F_X(x) = P(X \leq x)$$

#### Relation to PMF (Discrete Case):

$$F_X(x) = \sum_{t \leq x} p_X(t)$$

#### Relation to PDF (Continuous Case):

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

## Recovering the PDF from the CDF:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

### Properties of the CDF:

- $0 \leq F_X(x) \leq 1$
- $F_X(x)$  is non-decreasing
- $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $\lim_{x \rightarrow \infty} F_X(x) = 1$
- $F_X(x)$  is right-continuous

## Random Vector Random Vector and Joint Distribution

A random vector  $\mathbf{X} = (X_1, X_2, \dots, X_n)$  consists of  $n$  random variables defined on a common probability space. Each outcome is mapped to a point in the  $n$ -dimensional space  $R^n$ , allowing the combined behavior of multiple random variables to be described.

The joint cumulative distribution function (CDF) of  $\mathbf{X}$  is defined as

$$F_{\mathbf{X}}(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

### Marginalization Property

If  $\mathbf{X} = (X, Y)$  has a joint probability density function  $f_{X,Y}(x, y)$ , then the individual (marginal) density of  $X$  is obtained by integrating out  $Y$ :

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

Similarly, the marginal density of  $Y$  is

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

**Conditional Distributions** Similar to conditional probability, the conditional distribution of a random variable with respect to other can be found

*Discrete Case:*

$$P(Y = y \mid X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}, \quad p_X(x) > 0$$

*Continuous Case:*

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}, \quad f_X(x) > 0$$

## Independence of Random Variables

Random variables  $X$  and  $Y$  are independent if the joint distribution separates into the product of the marginals.

*Discrete Case:*

$$p_{X,Y}(x, y) = p_X(x) p_Y(y), \quad \forall x, y$$

*Continuous Case:*

$$f_{X,Y}(x, y) = f_X(x) f_Y(y), \quad \forall x, y$$

### Expectation

The expectation of the random variable gives the mean of the values taken by the random variable weighted by their probability giving sense of the most probable value.

$$E[X] = \sum_x x P(X = x) = \int_{-\infty}^{\infty} x f_X(x) dx$$

### Properties of Expectation

- **Linearity:**  $E[aX + bY] = aE[X] + bE[Y]$ , meaning expectation distributes over linear combinations.
- **Constant:**  $E[c] = c$ , so the expectation of a constant is the constant itself.
- **Independence:**  $E[XY] = E[X] E[Y]$  when  $X$  and  $Y$  are independent.

### Law of the Unconscious Statistician (LOTUS)

$$E[g(X)] = \sum_x g(x) P(X = x) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

### Variance

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

Variance quantifies the spread of values around the mean, indicating how far observations tend to deviate from the average.

### Standard Deviation

$$\sigma(X) = \sqrt{\text{Var}(X)}$$

The standard deviation provides a measure of dispersion in the same units as the random variable.

### Properties of Variance

- **Linear Transformation:**  $\text{Var}(aX + b) = a^2 \text{Var}(X)$ , showing that scaling affects variance while shifts do not.
- **Sum (Independent Case):**  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$  when  $X$  and  $Y$  are independent.
- **Sum (General Case):**  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$ , accounting for dependence between variables.

**Covariance** This gives the sense of how closely the two random variables move with each other. This measures only the first order interdependence of the random variables

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

### Properties of Covariance

- $\text{Cov}(X, X) = \text{Var}(X)$ , so the covariance of a variable with itself equals its variance.

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ , showing that covariance is symmetric.
- $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$ , meaning scaling affects covariance linearly while shifts do not.
- If  $X$  and  $Y$  are independent, then  $\text{Cov}(X, Y) = 0$ , although the converse need not be true.

**Correlation Coefficient** The value of covariance is not relative, so it's quite difficult to get hold of the magnitude of correlation between the two random variables. This is resolved by the correlation coefficient which makes use of the Cauchy Schwarz inequality to normalise the covariance.

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

### Properties of Correlation

- $-1 \leq \rho(X, Y) \leq 1$ , so correlation is always bounded.
- $|\rho(X, Y)| = 1$  if and only if  $Y = aX + b$  for some constants  $a$  and  $b$ , indicating perfect linear dependence.
- $\rho(X, Y) = 0$  implies that  $X$  and  $Y$  are uncorrelated, but not necessarily independent.

## 2 Kalman Filtering

Kalman Filtering is an RMSE estimator, which minimizes the error in the model's output and the actual output. This is mainly used in cases where measurements from sensors are received which contain some noise due to which they do not accurately depict the state of the system. Kalman Filtering makes an intelligent weighing of the predicted state (from the previous state value) and the measured state such that the error is minimized.

$$\hat{x}_{\text{RMSE}} = \arg \min_{\hat{x}} E[(X - \hat{x})^2]$$

Here,  $X$  is the original state and  $\hat{x}$  is the predicted state. The state in Kalman filtering is the set of variables we want as the output. It is denoted as  $x_k$  where  $k$  denotes the instant of time. It's an  $n \times 1$  matrix.

$$x_{k+1} = \Phi x_k + w_k \quad (1)$$

$\Phi$  is the state transition matrix ( $n \times n$ ) from the state  $x_k$  to  $x_{k+1}$  assuming the process is stationary.  $w_k$  ( $n \times 1$ ) is the process white noise which accounts for uncertainty. It has a known covariance.

The measurement process can be modeled as below

$$z_k = Hx_k + v_k \quad (2)$$

The measurement made does not include all the state variables that are being considered. For example, for a state space containing position and velocity, only position might be measured. The velocity might be derived from it.  $z_k$  is the actual measurement obtained at time

instant  $k$  ( $m \times 1$ ). The measurement matrix  $H$  provides the noiseless connection between the current state and the measured values.  $v_k$  is the measurement white noise used to model the uncertainties in the measurement process. The measurement and process noise are independent of each other.

The covariances of the two noise models are assumed stationary over time and are given by

$$Q = E[w_k w_k^T] \quad (3)$$

$$R = E[v_k v_k^T] \quad (4)$$

The mean squared error is given by

$$E[e_k e_k^T] = P_k \quad (5)$$

$$P_k = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \quad (6)$$

Here  $x_k$  is the actual state and  $\hat{x}_k$  is the predicted state. The RMSE error matrix  $P_k$  is called the model uncertainty ( $n \times n$ ).

Let  $\hat{x}'_k$  be the prior estimate of  $\hat{x}_k$  from the knowledge of the system. As mentioned earlier, the final prediction  $\hat{x}_k$  is weighed by the kalman gain  $K_k$  as given below

$$\hat{x}_k = \hat{x}'_k + K_k(z_k - H\hat{x}'_k) \quad (7)$$

Substituting 2 in this gives

$$\hat{x}_k = \hat{x}'_k + K_k(Hx_k + v_k - H\hat{x}'_k) \quad (8)$$

Substituting into 6 gives

$$P_k = E\left[\left((I - K_k H)(x_k - \hat{x}'_k) - K_k v_k\right)\left((I - K_k H)(x_k - \hat{x}'_k) - K_k v_k\right)^T\right] \quad (9)$$

At this point it is noted that  $x_k - \hat{x}'_k$  is the error of the prior estimate. It is uncorrelated with the measurement error, so it is rewritten as

$$P_k = (I - K_k H) E\left[(x_k - \hat{x}'_k)(x_k - \hat{x}'_k)^T\right] (I - K_k H)^T + K_k E[v_k v_k^T] K_k^T \quad (10)$$

Further simplification gives

$$P_k = (I - K_k H) P'_k (I - K_k H)^T + K_k R K_k^T \quad (11)$$

Where,  $P'_k$  is the prior estimate of  $P_k$ . The sum of diagonal elements of  $P_k$  gives the sum of RMSE errors associated with all the state variables. Therefore the objective is to minimize the trace of  $P_k$ .

$$\text{tr}(P_k) = \text{tr}(P'_k) - 2 \text{tr}(K_k H P'_k) + \text{tr}\left(K_k (H P'_k H^T + R) K_k^T\right) \quad (12)$$

The trace of  $P - k$  is first differentiated with respect to  $K_k$  and the result set to zero in order to find the conditions of this minimum.

$$\frac{d \text{tr}(P_k)}{d K_k} = -2(H P'_k)^T + 2K_k (H P'_k H^T + R) \quad (13)$$

This gives the value of Kalmann Gain  $K_k$  as

$$K_k = P'_k H^T \left( H P'_k H^T + R \right)^{-1} \quad (14)$$

Substituting the value of kalmann gain into 11 gives the RMSE error  $P_k$  as

$$P_k = P'_k - P'_k H^T \left( H P'_k H^T + R \right)^{-1} H P'_k \quad (15)$$

$$P_k = (I - K_k H) P'_k \quad (16)$$

The current state predictions which are projected from the previous state values is given by

$$\hat{x}'_k = \Phi \hat{x}_{k-1} \quad (17)$$

The uncertainty matrix  $P'_k$  is given by

$$P'_k = E \left[ e'_k (e'_k)^T \right] \quad (18)$$

$$e'_k = x_k - \hat{x}'_k = \Phi e_{k-1} + w_{k-1} \quad (19)$$

$$P'_k = E \left[ (\Phi e_{k-1} + w_{k-1}) (\Phi e_{k-1} + w_{k-1})^T \right] \quad (20)$$

$$P'_k = \Phi P_{k-1} \Phi^T + Q \quad (21)$$

So, in summary the following steps are taken in Kalmann filtering,

- Start with initial State matrix and initial uncertainty.
- Calculate the predicted uncertainty and state matrix using (21) and (17).
- Calculate the Kalmann Gain using (14)
- Obtain the measurement and update the estimate using (7)
- Report the uncertainty using (16)

### 3 1D single state Kalman Filtering

A kalmann filtering simulation for a state matrix containing a single variable, position was performed using python. The original trajectory was assumed to be moving with a constant velocity, but the measured value was added with white noise of variance R. The initial state and uncertainty was provided and the kalmann filtering steps were carried out in iteration to obtain the estimated trajectory. The impact of various parameters like the process uncertainty Q, Initial estimates, uncertainty was tested as explained below.

Base case with  $R = 4$ ,  $Q = 0.1$ ,  $P_{in} = 10$

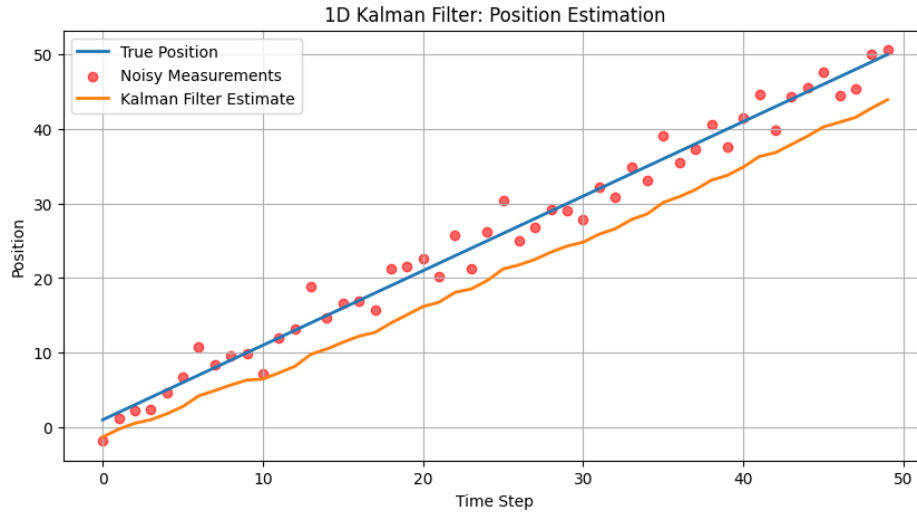


Figure 1: Base Case

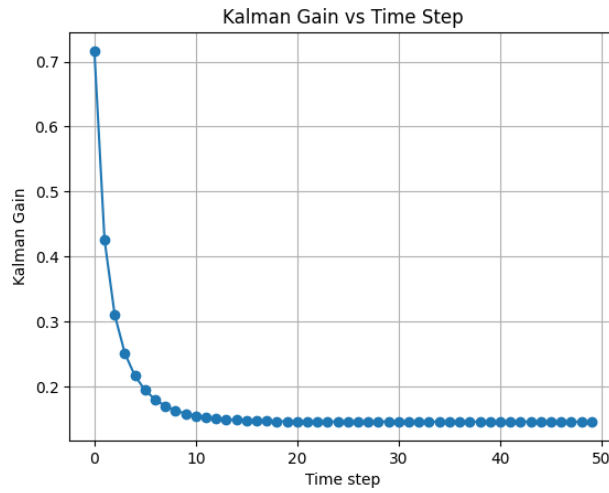


Figure 2: Kalmann Gain variation with time for base case

### Case 1 $R$ increased from 4 to 16

As a direct consequence of increasing  $R$ , the deviation of measurements from the original value has increased. This leads to an increased mismatch in the kalmann estimate with the true value but we could see the system adapting its kalmann gain in such a way to reduce dependency on the measured values faster than the previous case.



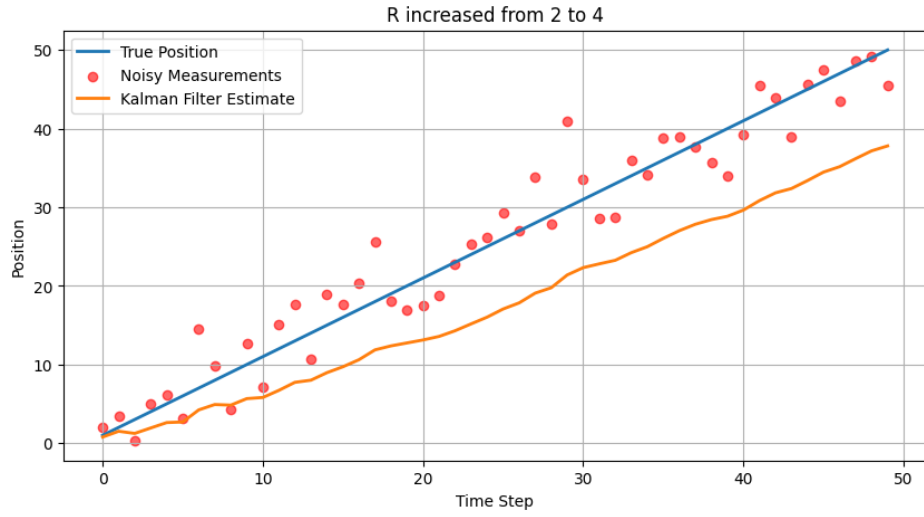


Figure 3: R increased from 4 to 16

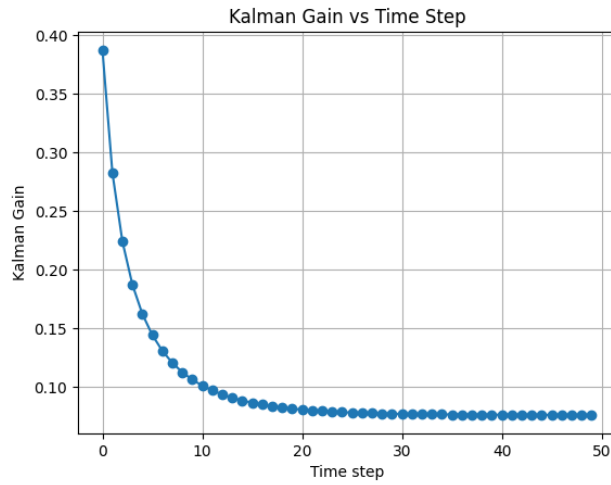


Figure 4: Kalmann Gain variation with time for case1

### Case 2 Q changed from 0.1 to 1

With the increase in process uncertainty the model estimate now follows the measured value closer than the base case. The model adapts itself to use the measured value with a higher weight as it settles the Kalmann gain at a higher value compared to the previous case.

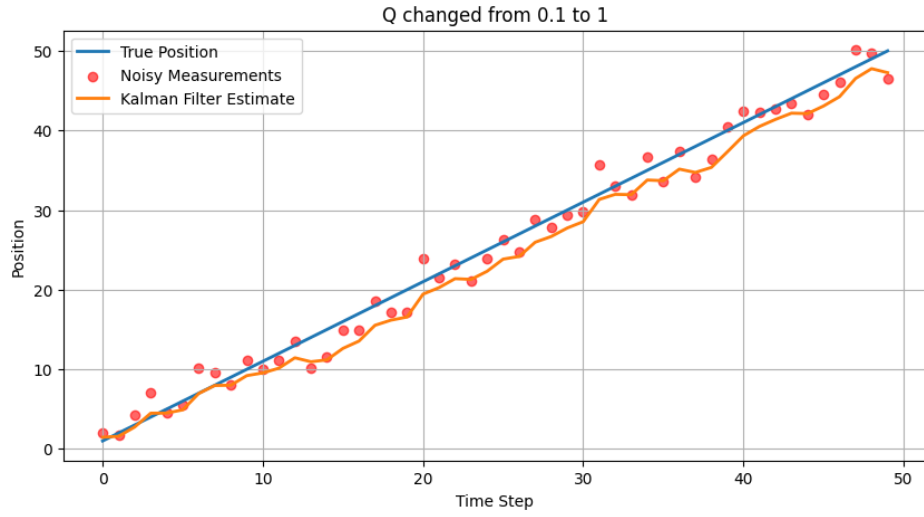


Figure 5: Q changed from 0.1 to 1

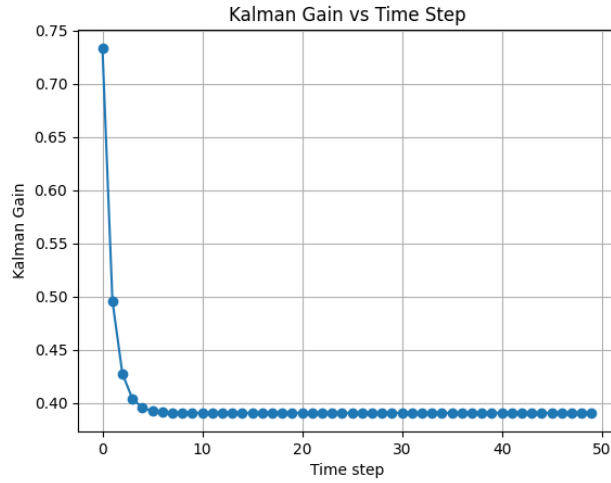


Figure 6: Kalmann Gain variation with time for case2

### Case 3 Reduced initial process uncertainty from 10 to 1

The effect of reduction of initial uncertainty is not much evident on the long run performance of the system. However initially the model closely follows the predicted values but finally it is self correcting and settles at a similar kalmann gain as the base case indicating its self sustaining ability.

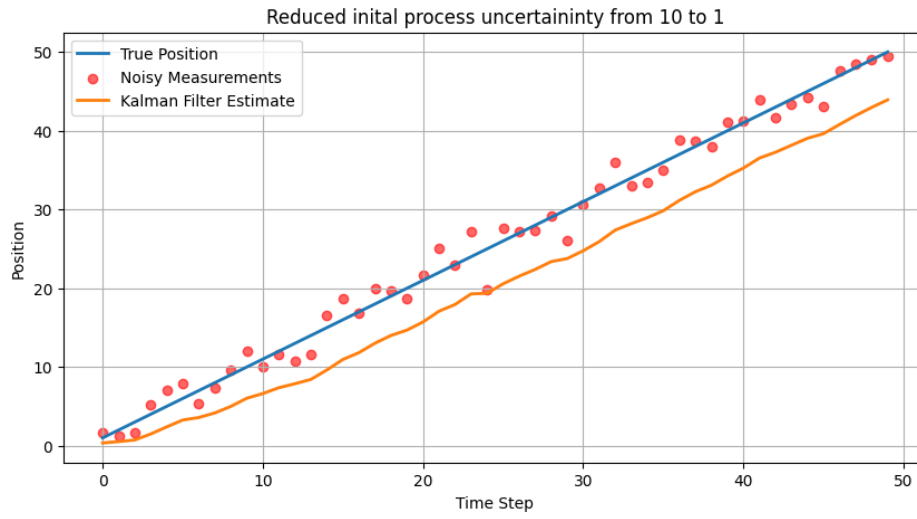


Figure 7: Reduced initial process uncertainty from 10 to 1

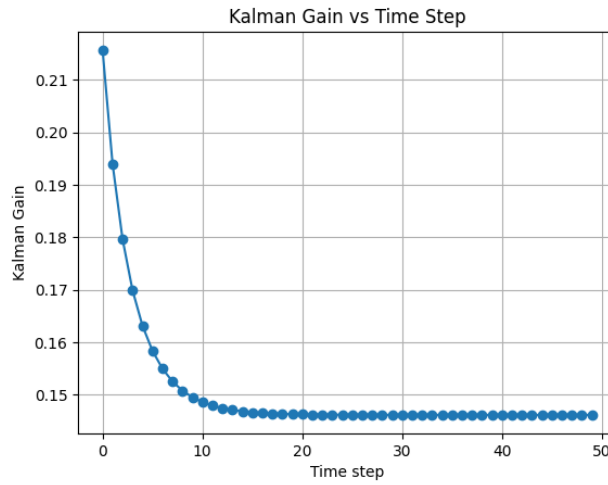


Figure 8: Kalmann Gain variation with time for case3

#### Case 4 Initial state changed from 0 to 100

Even if the initial measurement is off by a great value, the model is able to recover from it due to the large uncertainty associated with the initial guess. This shows the robustness of the kalman filter to navigate through an unknown initial state.

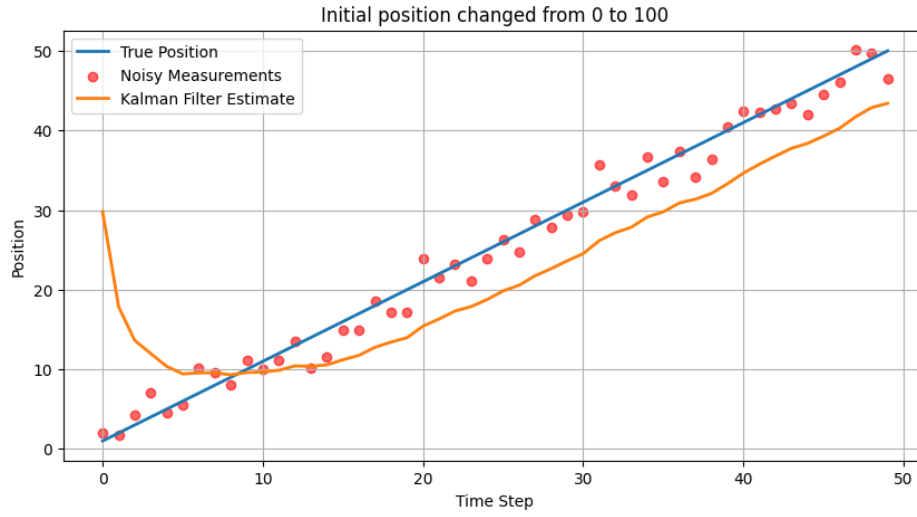


Figure 9: initial state changed from 0 to 100

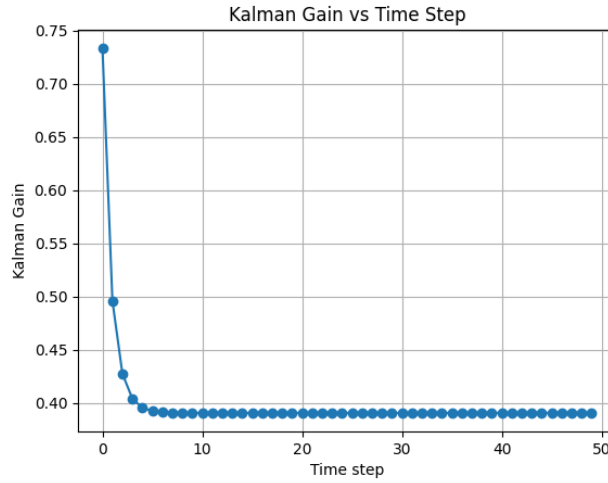


Figure 10: Kalmann Gain variation with time for case4

## 4 1D 2 State Estimation using Kalmann Filtering

Instead of just estimating the position the velocity of the object is also estimated in this simulation. Differing from the pervious case in which the object in study moves with constant velocity, now the object is moves with some acceleration which is added randomly to the velocity. This random acceleration is considered as the process noise and the state estimation is carried out and tuning of  $Q$  and  $R$  to minimize the prediction error is done.

**State vector**

$$\mathbf{x}_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}$$

**Initial estimate covariance**

$$\mathbf{P}_0 = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$$

State transition matrix

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

Measurement matrix (position only)

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Process noise covariance (random acceleration)

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & \Delta t^2 \end{bmatrix}$$

Measurement noise covariance

$$\mathbf{R} = \begin{bmatrix} \sigma_z^2 \end{bmatrix}$$

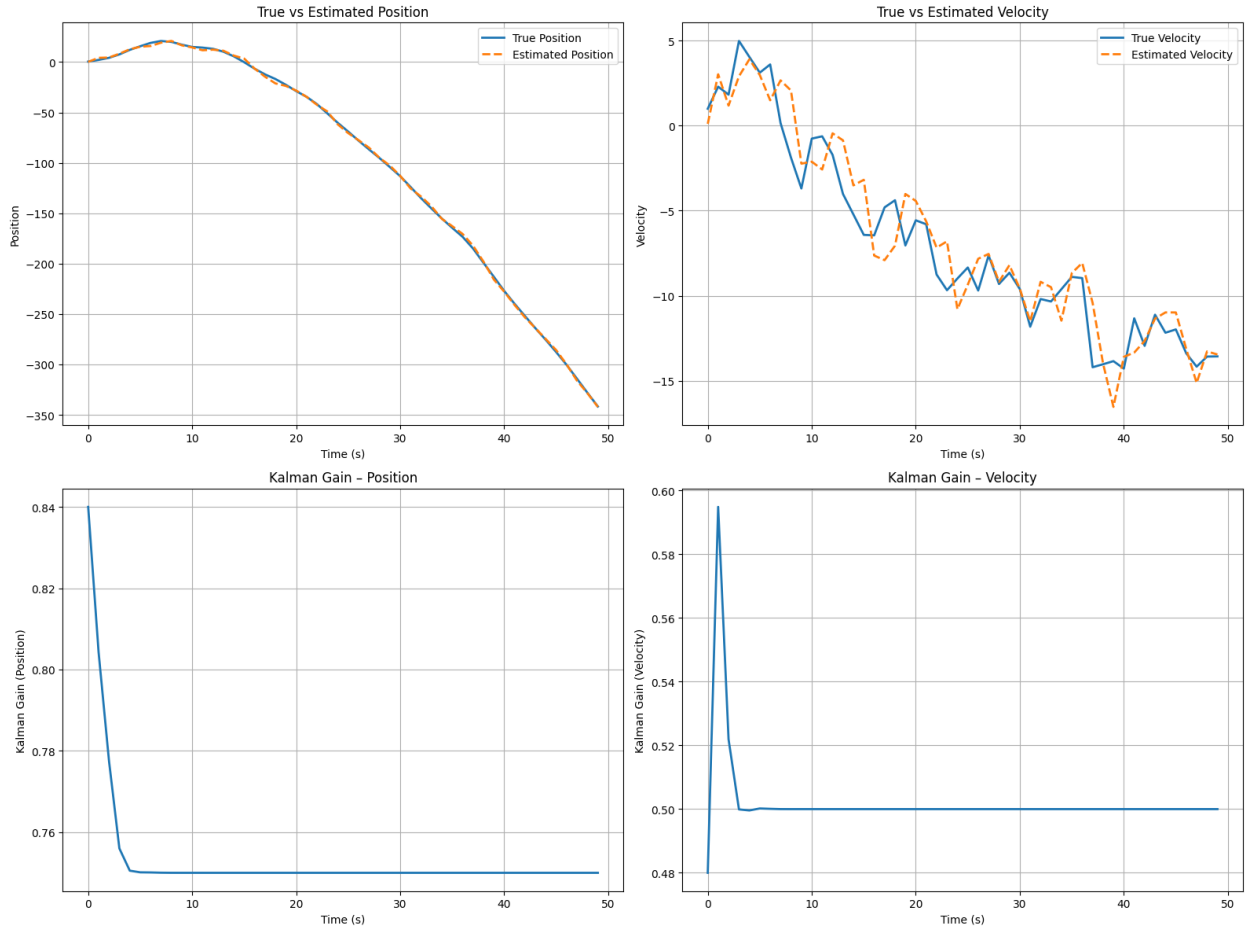


Figure 11: 2 State estimation

## 4.1 Tuning Q and R

Upon altering the time step we directly modify the process uncertainty, increasing by increasing the time step, and the measurement standard deviation is also swept from 0.5

to 5. The graphs indicate that there is no clear trend in the estimation error with  $Q$  and  $R$ . However a grid search can be performed on these values to find the set of values for which the estimation error is minimum.

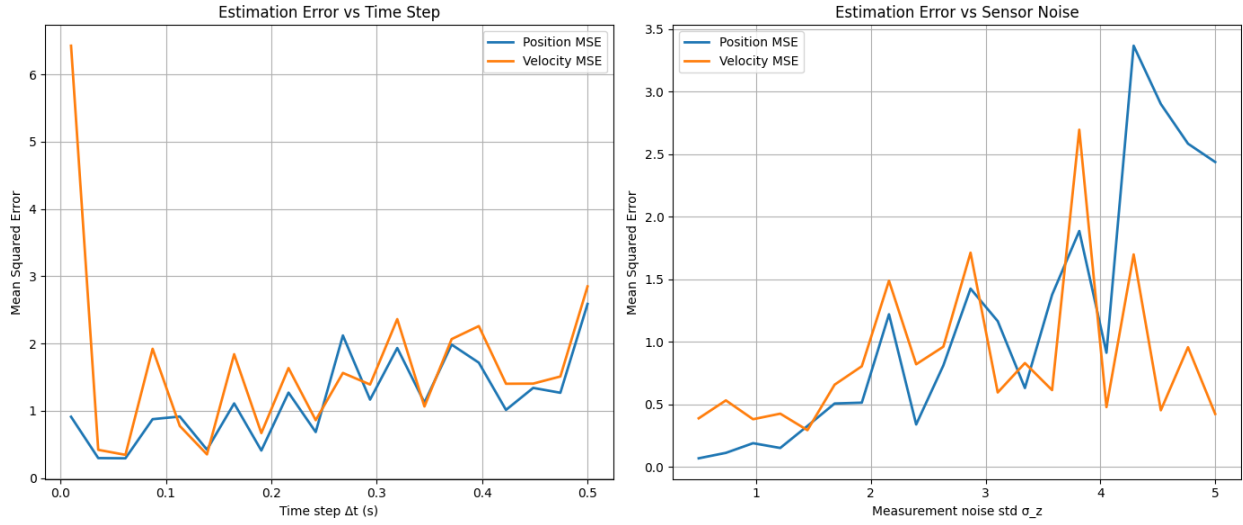


Figure 12: Dependence of Error on  $Q$  and  $R$  individually

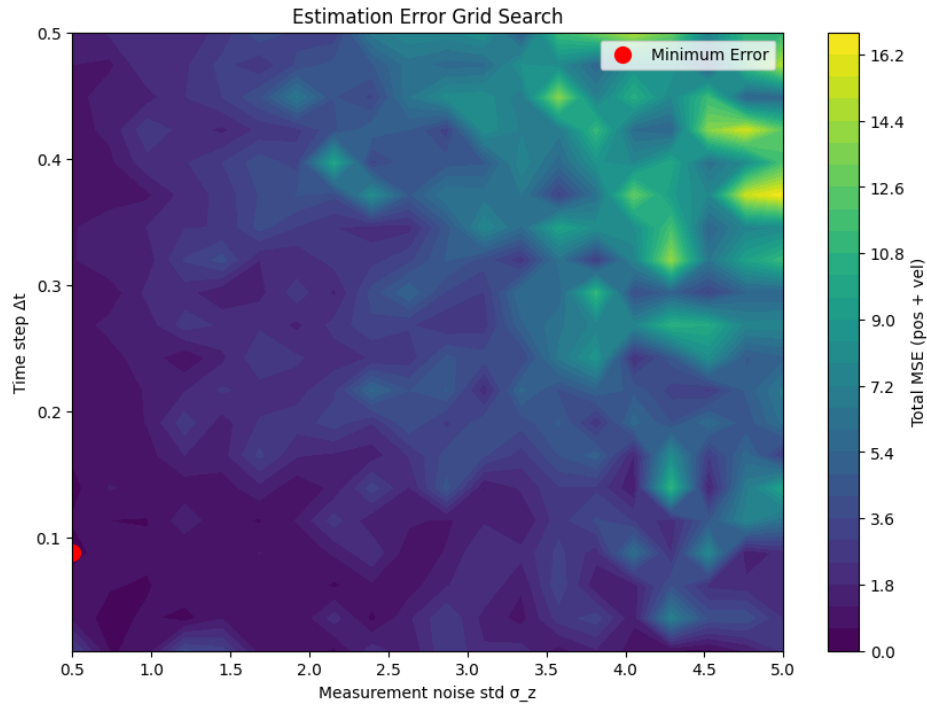


Figure 13: Grid Search for optimal  $Q$  and  $R$

**Minimum estimation error**

$$\varepsilon_{\min} = 0.268$$

**Optimal time step**

$$\Delta t^* = 0.087 \text{ s}$$

Optimal measurement standard deviation

$$\sigma_z^* = 0.500$$